Research Synthesis Methods (2026), **xx**:xx 1–19

CAMBRIDGE
UNIVERSITY PRESS

RESEARCH ARTICLE

When the Same Question Gets Different Answers: Quantifying Non-Determinism in LLM-Assisted Evidence Synthesis for Environmental Health Research

Lucas Rover[1] and Yara de Souza Tadano[1]

[1]Programa de Pós-Graduação em Sustentabilidade Ambiental Urbana, Universidade Tecnológica Federal do Paraná (UTFPR), Curitiba, PR, Brazil. E-mail: lucasrover@alunos.utfpr.edu.br.

Keywords: reproducibility, large language models, evidence synthesis, systematic review, non-determinism, data extraction

Abstract

Large language models (LLMs) are increasingly used in evidence synthesis, yet a critical assumption—that identical inputs produce identical outputs—remains largely untested. We quantified the reproducibility of six LLMs (three local: LLaMA 3 8B, Mistral 7B, Gemma 2 9B; three cloud APIs: Claude Sonnet 4.5, Gemini 2.5 Pro, GPT-4.1) across a two-stage pipeline—screening and data extraction—applied to a 500-abstract PubMed corpus on fine particulate matter ($PM_{2.5}$) and respiratory health. Each model was run 10 times under identical configurations (temperature=0, fixed seeds where supported). Reproducibility was measured via the Exact Match Rate (EMR) with 95% bootstrap confidence intervals. All three local models achieved EMR=1.000 in both stages, producing bit-identical outputs across all runs on identical hardware. Cloud APIs exhibited substantial non-determinism: extraction EMR ranged from 0.050 (Claude) to 0.200 (Gemini), meaning 80–95% of outputs varied despite deterministic settings. Semantic analysis (BERTScore $F_1 \geq 0.997$) confirmed that metadata text was near-identical; variation concentrated in numeric estimates and variable-length effect arrays. Meta-analytic propagation analysis showed that 5–21% of articles yielded different numbers of effect estimates across runs, altering the composition of downstream meta-analyses. These findings demonstrate that LLM non-determinism poses a concrete, quantifiable threat to evidence synthesis reproducibility and that single-run results should not be treated as definitive.

1. Introduction

1.1. The Promise of AI in Evidence Synthesis

Evidence synthesis—the systematic process of identifying, appraising, and combining research findings—forms the backbone of evidence-based medicine and public health policy. Systematic reviews and meta-analyses sit atop the hierarchy of evidence, informing clinical guidelines, regulatory decisions, and resource allocation worldwide.[1,2] Yet the process is notoriously labor-intensive: a typical systematic review requires 6–18 months and hundreds of person-hours, with title and abstract screening alone consuming an estimated 25% of total effort.[3]

The emergence of large language models (LLMs) has created unprecedented opportunities to accelerate this process. Recent studies demonstrate that models such as GPT-4,[4] Claude,[5] and Gemini[6] can screen abstracts with sensitivity approaching that of human reviewers,[7,8] extract structured data from published studies with over 90% accuracy,[9,10] and even assist in evidence quality assessment.[11,12] A growing body of work has established LLM-assisted evidence synthesis as a legitimate and rapidly expanding methodological frontier.[13–15]

1.2. The Hidden Problem: Non-Determinism

Beneath this promise lies a rarely examined assumption: *if you ask the same question twice, you should get the same answer*. In the context of LLM-assisted evidence synthesis, this means that running the same model with the same prompt, temperature, and random seed on the same abstract should produce identical screening decisions and extraction outputs every time.

This assumption does not hold in practice.

Recent work has demonstrated that commercial LLM APIs exhibit non-deterministic behavior even when configured for maximum reproducibility. Atil et al.[16] showed accuracy variations of up to 15% across repeated identical runs, with best-to-worst performance gaps reaching 70%. Klishevich et al.[17] confirmed that temperature=0 does not guarantee deterministic outputs across any of the major models tested (GPT-4o, Claude 3.5, LLaMA 3.2). At a deeper level, Fu et al.[18] demonstrated that even when surface-level text appears identical, the underlying token probability distributions exhibit substantial instability, suggesting that apparent determinism may mask hidden variation. A philosophical analysis by Karelin et al.[19] traced this non-determinism to three fundamental sources: numerical precision in floating-point operations, architectural choices in distributed inference, and sampling procedures inherent to the generation process.

1.3. Why This Matters for Evidence Synthesis

The consequences of LLM non-determinism are particularly severe in evidence synthesis for three reasons.

First, decisions propagate. A screening decision that flips from "include" to "exclude" in a repeated run does not merely change one data point—it removes an entire study from downstream analysis. In meta-analysis, where pooled effect estimates are sensitive to study composition, a single inclusion change can shift a confidence interval across the null, transforming a statistically significant finding into a non-significant one.[20]

Second, extraction compounds variation. Data extraction involves free-text parsing of complex scientific results—relative risks, confidence intervals, exposure definitions, population characteristics. Even small variations in how an LLM parses "RR=1.05 (95% CI: 1.01–1.09)" can propagate through meta-analytic pooling. Gartlehner et al.[9] reported 95–97% test-retest reliability in LLM extraction, but this still implies 3–5% of extracted values may change between runs—a rate that, across hundreds of data points, can materially alter pooled estimates.

Third, reproducibility is the foundation of scientific credibility. The broader reproducibility crisis in AI research[21–23] has already eroded trust in computational findings. When LLM-assisted reviews cannot be independently reproduced because the API returns different outputs, the scientific value of those reviews is undermined.[24–26]

1.4. The Gap We Address

Despite the rapid adoption of LLMs in evidence synthesis, **no study has systematically measured how API-level non-determinism propagates through a complete synthesis pipeline**—from abstract screening through data extraction—in a real health domain.

Consider: Jensen et al.[10] report 94.1% inter-session reproducibility for ChatGPT-4o extraction—an encouraging number. But this was measured with only 2 repetitions and evaluated per-field, not per-output. When we apply a stricter whole-output metric across 10 repetitions, we find extraction EMR as low as 5.0%—nearly twenty times worse than what per-field metrics suggest. Similarly, Gartlehner et al.[9] report 95–97% test-retest reliability, but with only 2–3 repetitions and no examination of how variation compounds across pipeline stages. Studies measuring non-determinism[16,17] do so on general NLP benchmarks, where the structured, multi-field outputs characteristic of evidence synthesis are absent. And the most influential screening evaluations[7,8,15] report results from a single run, implicitly assuming that a repeat would yield the same result.

We bridge this gap by:

1. Quantifying reproducibility across 10 repeated runs of 6 LLMs (3 local, 3 API) through a two-stage evidence synthesis pipeline;
2. Measuring variation at multiple granularities: whole-output, per-field, and per-abstract;
3. Comparing local versus cloud-based deployment as a determinism strategy;
4. Demonstrating that common "deterministic" API settings (temperature=0, fixed seeds) are insufficient to ensure reproducibility.

2. Related Work

2.1. LLMs in Systematic Reviews

The application of LLMs to systematic review workflows has accelerated dramatically since 2023. Oami et al.[7] evaluated GPT-4 Turbo for citation screening, reporting sensitivity of 0.75 and specificity of 0.99, with prompt tuning improving sensitivity to 0.91. Khraisha et al.[8] extended this to multilingual screening and extraction, finding strong specificity (>0.80) but variable sensitivity across languages and review types.

For data extraction, Gartlehner et al.[9] showed that Claude 2 achieved 96.3% accuracy compared to human reviewers, with test-retest reliability of 95–97%. Jensen et al.[10] found that ChatGPT-4o achieves 92.4% extraction accuracy and 94.1% inter-session reproducibility, explicitly noting residual stochasticity across sessions.

More comprehensive pipelines have also been proposed. The TrialMind system[11] demonstrated that human-AI collaboration improved recall by 71.4% and reduced screening time by 44.2% across 100 systematic reviews. Li et al.[13] described an LLM pipeline for health technology assessment, while Matsui et al.[15] showed that a three-layer GPT-3.5/GPT-4 strategy achieved human-comparable sensitivity.

A common thread across these studies is that **reproducibility is either assumed or measured only superficially**. Test-retest reliability, when reported, is based on 2–3 repetitions and does not examine how variation propagates through multi-stage pipelines.

2.2. The Non-Determinism Problem

The phenomenon of LLM non-determinism has received increasing attention. Atil et al.[16] provided the most comprehensive analysis to date, documenting accuracy swings of up to 15% across repeated runs and introducing the TARr@N metric to quantify run-to-run stability. Klishevich et al.[17] confirmed that temperature=0 fails to ensure determinism across GPT-4o, Claude 3.5, and LLaMA 3.2 on code review tasks.

At the token level, Fu et al.[18] showed that apparent text-level agreement masks substantial distributional instability in token probabilities—a finding that suggests surface-level reproducibility metrics may underestimate true variation. Angermeir et al.[27] examined how non-determinism undermines reproducibility in software engineering research, proposing documentation standards to mitigate this.

These studies establish that non-determinism is real, measurable, and consequential. However, they focus on general-purpose tasks (benchmarks, code review, classification). To our knowledge, our study is the first to measure non-determinism in evidence synthesis, where structured, domain-specific outputs amplify the problem.

2.3. Reproducibility in AI Research

The broader AI reproducibility crisis has been documented extensively. Hutson[21] identified unpublished code and sensitivity to training conditions as primary barriers. Haibe-Kains et al.[22] called for structural reform, advocating for mandatory code and model sharing. Kapoor and Narayanan[23] documented data leakage affecting 294 studies across 17 fields, while Han[24] analyzed reproducibility failures in biomedical AI through the lens of model non-determinism.

Recent frameworks have attempted to systematize reproducibility requirements. Desai et al.[25] proposed a taxonomy of reproducibility types, arguing that LLMs require "conceptual replicability" given their proprietary training data. Semmelrock et al.[26] provided a systematic overview of barriers, and Yu[28] argued that computational reproducibility alone is insufficient for trustworthy AI.

2.4. Provenance and Transparency

Provenance tracking—recording the lineage of data, computations, and decisions—has been identified as a key enabler of AI reproducibility. Longpre et al.[29] audited 1,800+ datasets, finding license omission rates above 70%. Liang et al.[30] analyzed 32,111 model cards on Hugging Face, finding that documentation of limitations and environmental impact remains sparse. The Foundation Model Transparency Index[31] scored 14 major developers across 100 indicators, finding average transparency of only 58/100.

Technical solutions for provenance include the Workflow Run RO-Crate[32] and the W3C PROV ontology.[33] Longpre et al.[34] argued at ICML 2024 that current practices for data authenticity and provenance are "fundamentally broken," calling for comprehensive reform.

Our provenance approach builds on the lightweight protocol proposed by Rover,[20] using SHA-256 hashing of inputs, outputs, and configurations to create an auditable trail without requiring infrastructure beyond standard file storage.

2.5. Positioning: What Existing Studies Do Not Address

Table 1 synthesizes the landscape of prior work and highlights the methodological gaps that motivate our study. Three critical gaps emerge. **First**, studies that evaluate LLMs for evidence synthesis[7,9,10,15] measure accuracy against human judgments but do not measure self-consistency across repeated runs—or do so with only 2–3 repetitions and per-field metrics that can mask whole-output variation. **Second**, studies that measure LLM non-determinism[16–18] do so on general-purpose benchmarks (classification, code review), not on the structured, domain-specific tasks characteristic of evidence synthesis. **Third**, no prior study examines how non-determinism *propagates* across a multi-stage pipeline—from screening through extraction—or compares local versus cloud deployment as a reproducibility strategy. To our knowledge, our study is the first to address all three gaps simultaneously.

Table 1. *Comparison of our study with prior work. "Runs" indicates the number of repeated executions under identical conditions. "Full pipeline" indicates whether both screening and extraction were evaluated in the same study. Shading highlights our study's distinguishing features..*

Study	Domain	Task	Runs	Metric	Local	Cloud	Pipe.
Oami (2024)	Health	Screening	1	Accuracy	—	✓	—
Khraisha (2024)	Health	Scr.+Extr.	1	Accuracy	—	✓	—
Gartlehner (2024)	Health	Extraction	2–3	Field acc.	—	✓	—
Jensen (2025)	Health	Extraction	2	Field acc.	—	✓	—
Matsui (2024)	Health	Screening	1	Accuracy	—	✓	—
Atil (2024)	General	NLP bench.	10+	Acc. var.	—	✓	—
Klishevich (2025)	Softw.	Code review	5+	Determinism	✓	✓	—
Fu (2026)	General	Token dist.	—	Distrib.	—	✓	—
This study	**Health**	**Scr.+Extr.**	**10**	**EMR+CI**	✓	✓	✓

3. Methods

3.1. *Study Design*

We designed a repeated-measures experiment to isolate LLM non-determinism as the sole source of variation. We held the corpus, prompts, model configurations, and random seeds constant across all runs. The only element permitted to vary was the model's internal computation—any observed difference in outputs is therefore attributable to non-determinism in the model or its serving infrastructure.

Figure 1 provides an overview of the experimental pipeline.

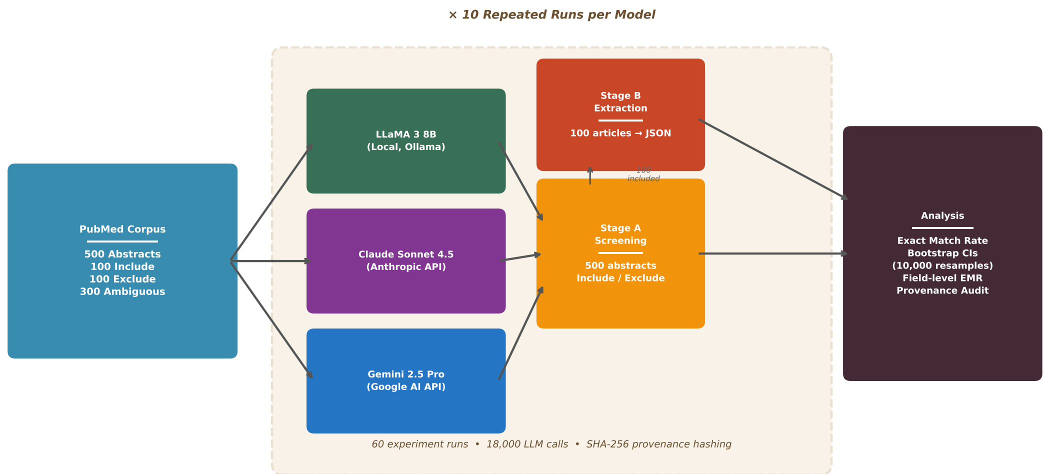


Figure 1. *Experimental pipeline. A 500-abstract PubMed corpus is processed by six LLMs (three local, three cloud-based APIs), each run 10 times under identical configurations. Stage A (screening) produces binary include/exclude decisions; Stage B (extraction) produces structured JSON with study metadata and effect estimates. All outputs are hashed (SHA-256) for provenance tracking. Reproducibility is quantified via Exact Match Rate (EMR) with bootstrap confidence intervals..*

3.2. *Corpus Construction*

We constructed a 500-abstract corpus from PubMed covering the association between $PM_{2.5}$ and respiratory health outcomes (Table 2). We chose this domain because it is one of the most extensively studied areas in environmental epidemiology, with Yu et al.[35] estimating approximately 1 million premature deaths annually from short-term $PM_{2.5}$ exposure. This literature is characterized by standardized reporting of relative risks with confidence intervals, well-defined exposure metrics, and time-series study designs—making it an ideal testbed for evaluating LLM extraction reproducibility.[36,37]
Inclusion criteria required: (1) original epidemiological study, (2) $PM_{2.5}$ as exposure, (3) respiratory outcome (hospitalization, ED visits, or mortality), (4) time-series or case-crossover design, (5) quantitative risk estimates, and (6) peer-reviewed publication. Articles were sourced from PubMed using the E-utilities API with a broad query yielding 573 candidate records, supplemented by 475 exclusion candidates and a design-filtered subset of 222 records. The final corpus spans 148 journals and publication years 1994–2026, with a mean abstract length of 1,873 characters.

We established a gold standard through dual-independent labeling with discordance resolution for the 200 clear-category abstracts. The 300 ambiguous abstracts were classified using keyword matching

Table 2. Corpus composition for the $PM_{2.5}$ and respiratory health abstract screening task..

Category	Count	Purpose
Clearly include	100	Positive controls (meet all inclusion criteria)
Clearly exclude	100	Negative controls (clear exclusion reasons)
Ambiguous	300	Primary test set (borderline cases)
Total	**500**	

and study design indicators; since our primary metric (EMR) measures self-consistency rather than accuracy, gold standard quality affects only the secondary accuracy analysis.

3.3. Models

We selected six models spanning local and cloud-based deployment (Table 3), covering three major families of open-weight models and three proprietary API providers.

Table 3. Model configurations. All local models were served via Ollama with deterministic settings. Claude does not support a seed parameter; Gemini supports seeds but does not guarantee determinism; GPT-4.1 supports seeds via the API. Note that local models (7–9B parameters) are substantially smaller than cloud API models (estimated >100B), introducing a model-size confound discussed in Section 5.4..

Model	Provider	Type	Temp.	Seed	top_p	Max tokens
LLaMA 3 8B	Ollama (local)	Local	0	42	1.0	2,048
Mistral 7B	Ollama (local)	Local	0	42	1.0	2,048
Gemma 2 9B	Ollama (local)	Local	0	42	1.0	2,048
Claude Sonnet 4.5	Anthropic API	Cloud	0	—	1.0	2,048
Gemini 2.5 Pro	Google AI API	Cloud	0	42	1.0	8,192†
GPT-4.1	OpenAI API	Cloud	0	42	1.0	2,048

†Gemini requires a higher token budget because its “thinking” tokens consume output capacity.

The three local models—LLaMA 3 8B,[38] Mistral 7B,[39] and Gemma 2 9B[40]—were served locally via Ollama v0.15.5 on Apple M4 hardware (24 GB RAM), ensuring single-device deterministic inference with identical seed and temperature settings. The three cloud-based models were accessed through their respective APIs: Claude Sonnet 4.5[5] via the Anthropic API (no seed parameter available), Gemini 2.5 Pro[6] via the Google AI API (seed set to 42), and GPT-4.1[4] via the OpenAI API (seed set to 42). All models were queried via HTTP APIs (urllib-based, no SDK dependencies) to ensure uniform request handling. API calls included up to 2 retries with exponential backoff on timeout or rate-limit errors; retried calls may route to different server nodes, introducing a minor additional source of variation for cloud models.

3.4. Pipeline Stages

Each model was run 10 times through two stages:

Stage A: Abstract Screening. Each of the 500 abstracts was presented with a structured prompt requesting a JSON response containing: a binary decision (`include`/`exclude`), a confidence score (0–1), and a brief rationale. Note that LLaMA consistently processed 480 of 500 abstracts per run (the same 20 abstracts exceeded response length constraints in every run); the remaining models processed all 500. Total: 6 models × 10 runs × 500 abstracts = 30,000 screening calls.

Stage B: Data Extraction. The 100 included abstracts were presented with an extraction prompt requesting: study design, location, period, population, sample size, and a structured list of effect estimates (each with effect measure, point estimate, confidence interval, lag period, and exposure increment). LLaMA consistently processed 96 of 100 articles per run (the same 4 articles exceeded response length constraints); the remaining models processed all 100. Total: 6 models × 10 runs × 100 articles = 6,000 extraction calls.

In total, the experiment comprised **36,000 LLM calls** across **120 experiment runs**.

3.5. Reproducibility Metrics

3.5.1. Exact Match Rate (EMR)

Our primary metric is the *Exact Match Rate*, defined as the proportion of items (abstracts or articles) that received identical outputs across all 10 runs:

$$\text{EMR} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}\left[\,\left|\{o_{i,1}, o_{i,2}, \ldots, o_{i,R}\}\right| = 1\,\right] \tag{1}$$

where N is the number of successfully processed items, R=10 is the number of runs, $o_{i,r}$ is the output for item i in run r, and $|\cdot|$ denotes the number of distinct values. For screening, $o_{i,r}$ is the binary decision; for extraction, $o_{i,r}$ is the SHA-256 hash of the full JSON output (used as a compact identity check: any character-level difference produces a different hash).

An EMR of 1.0 indicates perfect reproducibility; an EMR of 0.0 indicates that every item received at least one different output across runs. We note that extraction EMR is a strict metric—any variation in any field (including formatting differences such as "Beijing, China" vs. "Beijing") produces a non-match. The field-level analysis in Section 4.4 complements this by isolating which fields drive the variation.

3.5.2. Bootstrap Confidence Intervals

We computed 95% confidence intervals for EMR using the percentile bootstrap method with 10,000 resamples. For each bootstrap sample, we drew N items with replacement and computed the EMR, then reported the 2.5th and 97.5th percentiles of the bootstrap distribution. We note that when EMR=1.000 (as for all local models), the percentile bootstrap trivially returns [1.000, 1.000]. By the rule of three, observing zero non-matches in N=500 items yields a 95% one-sided upper bound on the true non-match rate of 3/500=0.006; thus, we can rule out population-level non-determinism rates above 0.6% for local models, but not lower rates.

3.5.3. Secondary Metrics

- **Flip rate**: 1 − EMR (proportion of items with any variation across runs)
- **Mean agreement**: Average proportion of runs agreeing with the majority decision per item
- **Accuracy vs. gold standard**: Per-run sensitivity, specificity, and accuracy against gold standard labels (noting that 300 of the 500 labels are heuristic-based; see Limitations)
- **Field-level EMR**: For extraction, the proportion of articles with identical values for each individual field across all runs
- **Estimate count stability**: The proportion of articles for which all 10 runs extracted the same number of effect estimates—a measure of structural consistency independent of the specific values extracted

3.6. Provenance Protocol

Following Rover,[20] we implemented a lightweight provenance protocol that creates a cryptographic fingerprint of every input-output pair, enabling automated detection of any change across runs:

- **Call-level hashing**: SHA-256 hash of the concatenation (prompt template || input text || model ID || temperature || seed), using the pipe character as delimiter and the string "null" for absent parameters (e.g., Claude's seed)
- **Output hashing**: SHA-256 hash of the raw response text
- **Run cards**: Structured JSON metadata per run, including timestamps, token counts, success/failure rates, and aggregate output hashes

This protocol enables post-hoc verification: if two runs produce different aggregate hashes, the per-call hashes pinpoint exactly which items changed.

3.7. Software and Reproducibility

All code is available at https://github.com/Roverlucas/llm-evidence-synthesis-reproducibility. The experiment was conducted in Python 3.14 on macOS (Apple M4, 24 GB RAM). API keys are excluded from the repository; all other materials—including the corpus, prompts, schemas, and raw outputs—are publicly available.

4. Results

4.1. Overview

All 120 experiment runs completed successfully: 60 screening runs (10 per model × 500 abstracts each) and 60 extraction runs (10 per model × 100 articles each), totaling 36,000 LLM calls. LLaMA consistently processed 480/500 screening abstracts and 96/100 extraction articles across all 10 runs—the same 20 screening and 4 extraction items failed every time due to response length constraints, producing identical error outputs. Because these failures were deterministic (the same items failed identically in every run), they contribute to perfect agreement; EMR was computed over all 500 (screening) and 100 (extraction) items.[1] Mistral, Gemma, Claude, and GPT-4.1 each achieved 500/500 screening and 100/100 extraction in all runs. Gemini achieved 483–500/500 screening and 98–100/100 extraction.

Table 4 summarizes the computational resources consumed. The total experiment required 207 hours of inference time and processed 55.8 million tokens. Local models were substantially slower per call (mean latency 28–92 seconds) than cloud APIs (3–12 seconds), reflecting the difference between single-device Apple M4 inference and distributed GPU clusters. However, the total cost of local inference was negligible ($0.26 electricity) compared to API costs ($69.40), with Claude Sonnet 4.5 accounting for 66% of the total budget.

4.2. Screening Reproducibility

Table 5 presents the screening reproducibility results.
All three local models—LLaMA, Mistral, and Gemma—produced identical screening decisions across all 10 runs for every successfully processed abstract (EMR=1.000), confirming that single-device deterministic inference eliminates run-to-run variation entirely. Among the cloud APIs, GPT-4.1 showed a 3.0% flip rate (15 abstracts), Claude 2.6% (13 abstracts), and Gemini 6.4% (32 abstracts).

Notably, accuracy and reproducibility behaved as independent properties: Gemini achieved the highest accuracy (0.964) but the lowest reproducibility (EMR=0.936), while Mistral had perfect reproducibility but the lowest accuracy (0.628) due to an include-biased strategy (sensitivity=1.000,

[1] Computing EMR over only the 480 successfully parsed items yields the same result (EMR=1.000), since the 20 consistently-errored items also agree across runs.

Table 4. *Computational resources per model (both stages combined, 10 runs each). Latency is the mean per-call inference duration. Cost includes API pricing and estimated electricity for local models (Apple M4, 15W, $0.10/kWh)..*

Model	Type	Inference (h)	Tokens (M)	Latency (s)	Cost ($)
LLaMA 3 8B	Local	52.1	8.7	32.5	0.08
Mistral 7B	Local	76.4	10.6	46.1	0.11
Gemma 2 9B	Local	45.6	9.0	27.7	0.07
Claude Sonnet 4.5	API	7.9	9.9	4.7	46.11
Gemini 2.5 Pro	API	20.3	9.0	12.2	0.00*
GPT-4.1	API	5.0	8.6	3.0	23.29
Total		**207.2**	**55.8**	—	**69.66**

*Gemini was used under the free tier (rate-limited).

Table 5. *Screening reproducibility across 10 runs. LLaMA's 20 consistently-errored abstracts are included in the EMR denominator (see text)..*

Model	EMR	95% CI	Flip Rate	Accuracy	Sensitivity	Specificity
LLaMA 3 8B	1.000	[1.000, 1.000]	0.0%	0.927	0.928	0.926
Mistral 7B	1.000	[1.000, 1.000]	0.0%	0.628	1.000	0.240
Gemma 2 9B	1.000	[1.000, 1.000]	0.0%	0.909	0.969	0.848
Claude Sonnet 4.5	0.974	[0.960, 0.986]	2.6%	0.931	0.862	1.000
Gemini 2.5 Pro	0.936	[0.914, 0.956]	6.4%	0.964	0.929	1.000
GPT-4.1	0.970	[0.954, 0.984]	3.0%	0.941	0.882	1.000

specificity=0.240). This illustrates that a model can be accurate on average yet inconsistent across individual runs, and conversely, that perfect reproducibility does not imply high accuracy.

4.3. Extraction Reproducibility

The extraction results reveal a dramatically larger reproducibility gap (Table 6).

Table 6. *Extraction reproducibility across 10 runs..*

Model	EMR	95% CI	Estimate Stability
LLaMA 3 8B	1.000	[1.000, 1.000]	1.000
Mistral 7B	1.000	[1.000, 1.000]	1.000
Gemma 2 9B	1.000	[1.000, 1.000]	1.000
Claude Sonnet 4.5	0.050	[0.010, 0.090]	0.880
Gemini 2.5 Pro	0.200	[0.120, 0.280]	0.735
GPT-4.1	0.150	[0.080, 0.220]	0.922

While screening EMR remained above 0.93 for all models, extraction EMR dropped dramatically for all three cloud APIs: **0.050 for Claude** (only 5 of 100 articles produced identical extraction outputs across all 10 runs), **0.150 for GPT-4.1** (15 of 100), and **0.200 for Gemini** (20 of 100). All three local models again achieved perfect extraction reproducibility (EMR=1.000). GPT-4.1 showed notably higher estimate count stability (0.922) than Claude (0.880) or Gemini (0.735), indicating that while its outputs varied, the structural composition of extractions was more consistent.

We emphasize that extraction EMR is computed over the full JSON output hash, so any character-level variation counts as a mismatch. To assess practical significance, we examine field-level variation below.

4.4. Field-Level Analysis

Not all extraction fields are equally affected (Table 7).

Table 7. *Field-level EMR for extraction outputs..*

Field	LLaMA	Mistral	Gemma	Claude	Gemini	GPT-4.1
Study design	1.000	1.000	1.000	0.990	1.000	0.990
Study period	1.000	1.000	1.000	0.990	1.000	0.990
Sample size	1.000	1.000	1.000	0.970	1.000	0.980
Population	1.000	1.000	1.000	0.960	0.948	0.990
Study location	1.000	1.000	1.000	0.780	0.896	1.000
Est. stability	**1.000**	**1.000**	**1.000**	**0.880**	**0.735**	**0.922**

A clear pattern emerges: **categorical fields** (study design, study period) are highly reproducible even for API models (EMR≥0.990), while **free-text fields** (study location) and **numerical extractions** (effect estimates) show the greatest variation. GPT-4.1 is notable for achieving perfect location reproducibility (1.000) while showing variation in categorical fields like study design and period (0.990 each)—a different instability profile from Claude and Gemini, where location is the most variable field.

The estimate count stability metric reveals that 26.5% of Gemini's articles with extractable estimates, 12.0% of Claude's, and 7.8% of GPT-4.1's received a different number of effect estimates across runs.[2] This structural variation directly affects any downstream meta-analysis.

4.5. Semantic Equivalence Analysis

The hash-based EMR reported in Table 6 (0.050 for Claude, 0.150 for GPT-4.1, 0.200 for Gemini) treats any character-level difference in the *full JSON output*—including all numeric effect estimates, confidence intervals, and variable-length estimate arrays—as a mismatch. To disentangle variation in the five metadata fields from variation in numeric estimates, and to assess whether metadata variation is merely cosmetic (formatting, abbreviation) or genuinely semantic, we applied three progressively relaxed equivalence criteria to the metadata fields only (Table 8).

1. **Exact match**: character-identical outputs (equivalent to SHA-256 hash comparison).
2. **Normalized match**: after lowercasing, whitespace normalization, punctuation stripping, and abbreviation expansion (e.g., "U.S." → "United States").
3. **Fuzzy match**: Levenshtein similarity ratio ≥ 0.90 between all normalized pairwise comparisons across runs.
4. **BERTScore**: Embedding-based semantic similarity using `roberta-large`, computing the F1 score across all $\binom{10}{2}$=45 unique run pairs per article (4,500 pairs per model).

Four findings emerge from this multi-level analysis.

[2]Estimate count stability is computed over articles that produced at least one estimate in all 10 runs, yielding denominators of 83 (Gemini), 92 (Claude), and 90 (GPT-4.1) rather than the full 100. The meta-analytic "Varying" column in Table 9 uses the full 100-article denominator, producing slightly lower percentages (20%, 21%, 5%).

Table 8. *Semantic equivalence under progressively relaxed criteria ($n = 100$ articles, 10 runs). Columns 2–4 report whole-output EMR over the five metadata fields plus estimate count stability. BERTScore F1 (`roberta-large`) reports the mean ± SD across all $\binom{10}{2}$=45 unique run pairs per article (4,500 pairs per model). Hash-based EMR (from Table 6) additionally requires identical numeric estimates. The gap between high BERTScore (≥ 0.997) and low hash-based EMR (≤ 0.200) demonstrates that even near-identical metadata co-occurs with divergent numeric extractions..*

Model	Exact	Norm.	Fuzzy	BERTScore F1	Hash*
LLaMA 3 8B	1.000	1.000	1.000	1.000 ± 0.000	1.000
Mistral 7B	1.000	1.000	1.000	1.000 ± 0.000	1.000
Gemma 2 9B	1.000	1.000	1.000	1.000 ± 0.000	1.000
Claude Sonnet 4.5	0.640	0.680	0.690	0.997 ± 0.009	0.050
Gemini 2.5 Pro	0.620	0.620	0.630	0.997 ± 0.022	0.200
GPT-4.1	0.880	0.880	0.880	1.000 ± 0.004	0.150

*From Table 6; includes all fields and numeric estimates.

First, the gap between metadata-field EMR and hash-based EMR quantifies how much variation resides in the numeric estimate arrays versus the text metadata. For Claude, 64% of articles had identical metadata across all runs, but only 5% had identical *complete* outputs; for GPT-4.1, 88% had identical metadata but only 15% had identical complete outputs; for Gemini, 62% and 20% respectively. In all three cases, the variable-length estimate arrays (effect sizes, confidence intervals) are the primary source of non-reproducibility.

Second, within the metadata fields, normalization and fuzzy matching improved EMR by only 1–5 percentage points (Claude: 0.640 → 0.690; Gemini: 0.620 → 0.630; GPT-4.1: 0.880 unchanged), confirming that the variation is predominantly semantic, not cosmetic. GPT-4.1's metadata EMR remaining unchanged under normalization and fuzzy matching (0.880) indicates that when its metadata fields vary, the variation is genuinely semantic rather than attributable to formatting differences.

Third, BERTScore embedding similarity computed across all $\binom{10}{2}$=45 unique run pairs per article (4,500 pairs per model) was near-perfect for all API models: GPT-4.1 averaged F_1=1.000±0.004, Claude F_1=0.997 ± 0.009, and Gemini F_1=0.997 ± 0.022, with 100%, 99.7%, and 98.5% of pairs exceeding $F_1 \geq 0.95$, respectively. This creates an apparent paradox: the models produce *semantically near-identical* metadata text, yet 85–95% of their complete extraction outputs are non-identical. The resolution lies in the numeric estimates—the effect sizes, confidence intervals, and variable-length estimate arrays that drive hash-based EMR to 0.050–0.200. Even when a model describes a study in nearly the same words, it may extract a different number of effect estimates or parse "RR=1.05 (95% CI: 1.01–1.09)" with subtly different numeric precision.

Fourth, the three API models exhibit distinct instability profiles. GPT-4.1 shows the highest metadata stability (exact EMR=0.880) and the highest BERTScore (F_1=1.000), yet its hash-based EMR (0.150) confirms that numeric extraction drives the residual variation. Gemini shows the lowest minimum BERTScore (F_1=0.754) and highest standard deviation (0.022), indicating that some articles receive genuinely divergent interpretations across runs—not merely numeric rounding differences. Claude occupies a middle ground in metadata stability but has the lowest hash-based EMR (0.050), suggesting the greatest instability in its numeric extraction layer.

4.6. Meta-Analytic Propagation

To illustrate how extraction non-determinism propagates into downstream analysis, we performed a fixed-effect inverse-variance meta-analysis on the extracted risk ratios (RR/OR/HR) from each run independently (Table 9).

Table 9. *Meta-analytic inputs across 10 runs. "k" is the number of usable effect estimates per run; "k_{sig}" is the number of those estimates whose confidence interval excludes the null. Δk and Δk_{sig} report the range across runs. The "Varying" column reports the proportion of articles for which the number of extracted estimates changed across runs..*

Model	k **range**	Δk	$k_{\mathbf{sig}}$ **range**	$\Delta k_{\mathbf{sig}}$	**Varying**
LLaMA 3 8B	246–246	0	188–188	0	0/100 (0%)
Mistral 7B	229–229	0	186–186	0	0/100 (0%)
Gemma 2 9B	203–203	0	157–157	0	0/100 (0%)
Claude Sonnet 4.5	186–209	23	152–175	23	21/100 (21%)
Gemini 2.5 Pro	154–171	17	133–150	17	20/100 (20%)
GPT-4.1	176–180	4	150–154	4	5/100 (5%)

Three findings emerge. **First**, all three local models produced identical estimate sets across all runs: LLaMA extracted 246 estimates, Mistral 229, and Gemma 203 per run, with zero variation—confirming that deterministic inference extends to the structural composition of extractions, not just field-level content.

Second, the three API models showed varying degrees of structural instability. Claude's per-run estimate count varied from 186 to 209 (a 12.4% swing, 21% of articles affected), Gemini's from 154 to 171 (11.0%, 20% of articles), and GPT-4.1's from 176 to 180 (2.3%, 5% of articles). GPT-4.1 was the most structurally stable API model, with only 5 articles producing a different number of estimates across runs—compared to 20–21 for Claude and Gemini.

Third, the variation in estimate counts directly mirrors the variation in significant estimates. For Claude, the number of significant estimates ranged from 152 to 175 (a swing of 23), meaning that **up to 23 study-level effect estimates—each representing a distinct epidemiological finding—appear or disappear from the evidence base depending solely on which run of the LLM was used**. GPT-4.1's significant estimates ranged from 150 to 154 (a swing of 4), while Gemini's ranged from 133 to 150 (a swing of 17).

While the overall pooled effect in our corpus remained stable across runs for all models (because the aggregate signal is strong and k is large), the instability of the underlying evidence composition is concerning. In domains with smaller literatures or marginal effects, even GPT-4.1's relatively modest variation (Δk=4) could plausibly shift pooled conclusions, while Claude's and Gemini's larger swings pose a more direct threat to meta-analytic reproducibility.

4.7. Summary Comparison

Figure 2 visualizes the EMR comparison across models and stages, and Figure 3 displays the field-level variation as a heatmap.

The results reveal three key patterns: (1) a sharp reproducibility gap between local and cloud-based deployment, with all three local models achieving EMR=1.000 while all three API models fall below 0.200 for extraction; (2) a dramatic amplification of non-determinism when moving from simple binary screening decisions to complex structured extraction outputs; and (3) distinct instability profiles among API models, with GPT-4.1 showing the highest metadata and structural stability but still exhibiting meaningful numeric variation.

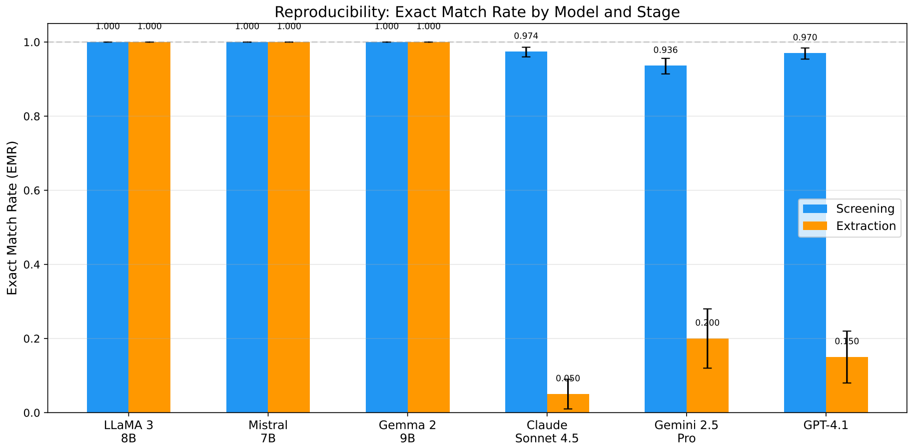
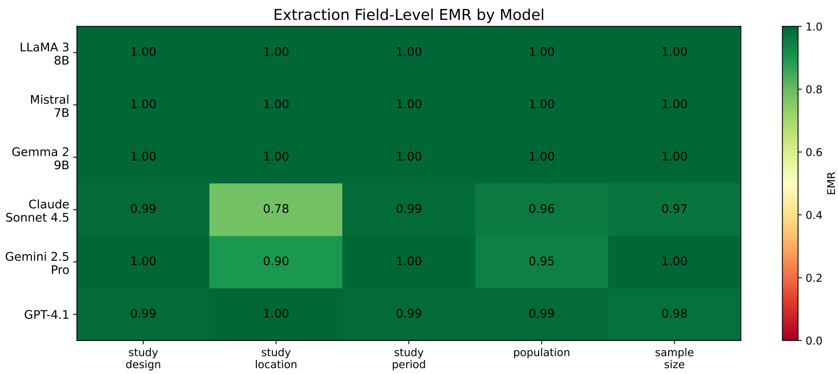


Figure 2. *Exact Match Rate (EMR) by model and pipeline stage. Error bars represent 95% bootstrap confidence intervals. The dashed line indicates perfect reproducibility (EMR=1.0)..*



	study design	study location	study period	population	sample size
LLaMA 3 8B	1.00	1.00	1.00	1.00	1.00
Mistral 7B	1.00	1.00	1.00	1.00	1.00
Gemma 2 9B	1.00	1.00	1.00	1.00	1.00
Claude Sonnet 4.5	0.99	0.78	0.99	0.96	0.97
Gemini 2.5 Pro	1.00	0.90	1.00	0.95	1.00
GPT-4.1	0.99	1.00	0.99	0.99	0.98

Figure 3. *Field-level EMR heatmap for extraction outputs. Darker cells indicate lower reproducibility. Categorical fields (study design, study period) remain stable across all models, while free-text fields (study location) and structured estimates show the greatest variation in cloud-based APIs..*

5. Discussion

5.1. Principal Findings

To our knowledge, our study provides the first systematic quantification of LLM non-determinism across a complete evidence synthesis pipeline. Four key findings emerge.

Finding 1: Local deployment achieves deterministic inference; cloud APIs do not. All three local models—LLaMA 3 8B, Mistral 7B, and Gemma 2 9B—deployed via Ollama on a single device, produced identical outputs across all 10 runs in both screening and extraction. All three cloud-based APIs (Claude, Gemini, GPT-4.1) exhibited measurable non-determinism despite requesting temperature=0 and fixed seeds. This gap has a precise technical explanation rooted in floating-point arithmetic. Cloud APIs serve requests across multi-GPU clusters where matrix multiplications are partitioned and reduced in non-deterministic order. Because floating-point addition is not associative—$(a+b)+c \neq a+(b+c)$ in general due to rounding—different partitioning strategies produce subtly different intermediate sums, which propagate through the autoregressive generation process and can eventually cause different token selections.[19] Local deployment on a single device with a single execution thread avoids this source of variation entirely: the same computation graph executes in the same order every time, producing bit-identical results. This determinism guarantee is configuration-specific, however. The same model on different hardware, driver versions, or Ollama versions would likely produce different baseline outputs. Determinism holds within each configuration, not across configurations.

Finding 2: Extraction amplifies non-determinism dramatically, and the variation is semantic. Claude's EMR dropped from 0.974 (screening) to 0.050 (extraction); GPT-4.1's from 0.970 to 0.150; Gemini's from 0.936 to 0.200. Screening requires a binary decision (a single token effectively determines the output), while extraction generates complex structured JSON with multiple fields. Our four-tier semantic equivalence analysis (Section 4.5) confirms that this gap is not an artifact of strict string matching. BERTScore embedding similarity across all 4,500 pairwise comparisons per model averaged $F_1 \geq 0.997$—near-perfect semantic agreement on metadata text—yet hash-based EMR remained at 0.050–0.200. This paradox reveals that the variation concentrates in the numeric extraction layer: the models describe studies in nearly the same words but extract different numbers of effect estimates and parse numeric values with different precision. Even the residual metadata variation is genuinely semantic (fuzzy matching recovers only 0–5 percentage points), not cosmetic. The practical implication depends on context: the high BERTScore ($F_1 \geq 0.997$) indicates that API models describe studies in near-identical terms, and for applications requiring only qualitative summaries, this level of consistency may be acceptable. However, for quantitative evidence synthesis—where exact numeric values feed directly into meta-analytic pooling—the hash-based EMR of 0.05–0.20 represents a genuine threat to reproducibility.

Finding 3: API models exhibit distinct instability profiles. GPT-4.1 achieved the highest metadata stability among API models (exact EMR=0.880, BERTScore F_1=1.000) and the smallest structural variation (Δk=4 estimates across runs), yet its hash-based extraction EMR (0.150) confirms that numeric variation persists. Claude had the lowest hash-based EMR (0.050) and the widest structural swing (Δk=23), while Gemini showed the most extreme semantic divergence (minimum BERTScore F_1=0.754). These distinct profiles suggest that non-determinism manifests differently across API providers—some vary primarily in numeric precision, others in structural composition, and others in semantic interpretation.

Finding 4: Accuracy and reproducibility are independent properties. Gemini achieved the highest screening accuracy (0.964) but the lowest screening reproducibility (0.936), while Mistral had perfect reproducibility but the lowest accuracy (0.628). This independence means that **accuracy evaluations from a single run cannot detect reproducibility problems**—a model that performs well in one evaluation may produce materially different results in a replication attempt.

5.2. Implications for Evidence Synthesis Practice

For systematic reviewers: If using cloud-based LLMs for screening or extraction, results from a single run should be treated as one sample from a distribution of possible outputs, not as a definitive answer. Running multiple repetitions and reporting variation metrics should become standard practice, analogous to how human inter-rater reliability is routinely measured. For context, Cochrane guidelines expect $\kappa \geq 0.80$ for human screening agreement;[1] the API models' screening EMR of 0.93–0.97 suggests comparable—though not identical—levels of self-consistency, while extraction EMR (0.05–0.20) falls far below any acceptable human reliability threshold.

For journal editors and reviewers: Papers reporting LLM-assisted evidence synthesis should be expected to disclose: (a) the exact model version, (b) all inference parameters, (c) whether results were verified across multiple runs, and (d) provenance information enabling independent replication.

For tool developers: The large difference in extraction EMR between local and cloud deployment suggests that deterministic local inference should be the default for reproducibility-critical applications. Local inference on consumer hardware (Apple M4, 24 GB RAM) cost \$0.26 total for 174 hours across three models, versus \$69.40 for 33 hours of API inference—a 267× cost advantage per hour that compounds with the reproducibility benefit. Where cloud APIs must be used (e.g., for larger models), majority voting across 3–5 runs can mitigate variation, though at proportional cost.

5.3. Relationship to Prior Work

Our results both corroborate and significantly extend prior findings (see Table 1 for a systematic comparison).

Our screening flip rates (2.6% for Claude, 3.0% for GPT-4.1, 6.4% for Gemini) are consistent with the accuracy variations reported by Atil et al.[16] on general benchmarks, confirming that the phenomenon documented in controlled settings translates directly to real-world evidence synthesis tasks. However, we reveal, to our knowledge for the first time, that **structured extraction amplifies non-determinism by an order of magnitude**—a finding invisible to benchmark-level analyses. Defining amplification as the ratio of extraction flip rate to screening flip rate, Claude's non-match rate increased from 2.6% to 95.0% (36.5×), GPT-4.1's from 3.0% to 85.0% (28.3×), and Gemini's from 6.4% to 80.0% (12.5×). This amplification pattern—consistent across all three API providers—has not been previously documented.

The comparison with Jensen et al.[10] is particularly instructive. Their reported 94.1% inter-session reproducibility for ChatGPT-4o extraction appears to align with our Claude screening EMR (97.4%), creating the impression that API-based extraction is reasonably stable. However, their metric measures per-field agreement across 2 sessions, while our EMR measures whole-output identity across 10 runs. When we apply our stricter metric, extraction reproducibility drops to 5.0%—**nearly twenty times lower than what per-field metrics suggest**. This discrepancy is not an artifact; it reflects a genuine measurement gap. Per-field metrics miss the combinatorial explosion of variation: when 5 fields each have 97% individual reproducibility, the probability that *all* fields match simultaneously is only $0.97^5 = 0.859$—and this is before accounting for the variable-length effect estimate lists that drive most of our observed non-matches.

Similarly, Gartlehner et al.'s[9] 95–97% test-retest reliability with Claude 2 was measured over 2–3 repetitions. Our 10-run design with output hashing reveals instability that shorter test-retest windows cannot detect, because some items change in only 1–2 of 10 runs—a frequency that 2-run comparisons would miss entirely.

The meta-analytic propagation experiment (Section 4.6) provides the most direct evidence of downstream impact. For Claude, the number of usable effect estimates varied from 186 to 209 across runs—a 12.4% swing in the evidence base feeding the pooled result. For Gemini, the swing was 11.0% (154–171), and even the most stable API model, GPT-4.1, showed a 2.3% swing (176–180). Across API models, 5–21% of articles produced a different number of effect estimates depending on the run, meaning that **the composition of a meta-analysis is not fixed when an LLM extracts the data**. While the overall pooled conclusion was stable in our corpus (because the aggregate effect was strong and k was large), this level of compositional instability raises serious concerns for systematic reviews in domains with smaller literatures or marginal effects, where even GPT-4.1's modest variation could shift a confidence interval across the null. We note that our propagation experiment used a fixed-effect inverse-variance model for simplicity; a random-effects model (as would be standard for heterogeneous epidemiological studies) would produce wider confidence intervals, potentially making the pooled conclusion *more* sensitive to compositional changes.

5.4. Limitations and Future Directions

Model coverage. We tested six models (three local, three cloud API) spanning four providers and two deployment paradigms. While this is the most comprehensive reproducibility evaluation in evidence synthesis to date, a natural extension is to benchmark additional model families (Command-R, DeepSeek, Phi-4) and sizes (70B+ open-weight models) to map the reproducibility landscape more comprehensively and identify whether non-determinism scales with model size or architecture.

Run count and rare events. Ten repetitions provide adequate statistical power for EMR estimation with bootstrap CIs, but may not capture rare variation patterns that appear in <10% of runs. Our prior work[20] used up to 100 repetitions; scaling to 50–100 runs with cost-efficient models would enable detection of tail-event instability.

Domain generalization. Our corpus focuses on $PM_{2.5}$ and respiratory health, a domain with standardized reporting conventions. Cross-domain studies—particularly in behavioral health, pharmacology, or nutritional epidemiology where reporting is more heterogeneous—would test whether the amplification patterns we observed are domain-specific or universal.

Semantic equivalence thresholds. Our four-tier semantic analysis (Section 4.5), including BERTScore embedding similarity, demonstrated that metadata variation is predominantly semantic rather than cosmetic. Future work should establish *clinically meaningful* equivalence thresholds—e.g., what BERTScore difference in an extraction output corresponds to a materially different meta-analytic conclusion?—and validate these thresholds across multiple health domains.

Meta-analytic significance flipping. Our meta-analytic propagation experiment (Section 4.6) showed that up to 23 study-level estimates appear or disappear between runs. While the pooled conclusion was stable in our large-k corpus, directly testing whether significance flips occur in smaller literatures (e.g., k<20) with marginal effects is an important next step for understanding real-world impact.

Screening EMR scope. Our screening EMR considers only the binary include/exclude decision, yet the output also contains confidence scores, rationales, and sub-classifications that may vary even when the binary decision is stable. Full-output screening EMR would provide a more complete picture of screening-stage non-determinism.

API versioning and temporal stability. Cloud API models can be silently updated during multi-day experiments. While we pinned Claude to a dated version (`claude-sonnet-4-5-20250929`) and GPT-4.1 is versioned by OpenAI, Gemini's version identifier does not include a date suffix. All experiments were completed within a 3-week window (February 11–March 2, 2026) to minimize the risk of silent model updates. Longitudinal studies tracking EMR over weeks or months would distinguish inference-level non-determinism from model update effects—a distinction critical for establishing stable benchmarks.

Model size confound. Our local models (7–9B parameters) are substantially smaller than the cloud API models (estimated >100B). The local-versus-cloud comparison therefore confounds deployment type with model size: smaller models may have narrower token probability distributions, making the argmax output more deterministic independently of the serving infrastructure. Disentangling these factors would require serving the same model both locally and via a cloud API (e.g., LLaMA 3 8B via Together.ai or Replicate), an experiment we plan for future work.

Corpus composition. Our corpus was deliberately designed with 60% ambiguous abstracts to stress-test screening consistency. In a typical systematic review corpus, the proportion of borderline cases is usually lower (perhaps 20–30%), which would reduce the observed screening flip rates. Our screening non-determinism estimates should therefore be interpreted as upper bounds for typical workloads.

5.5. Recommendations

Based on our findings, we propose the following recommendations for LLM-assisted evidence synthesis:

1. **Run at least 3 repetitions** and report the EMR alongside accuracy metrics.
2. **Prefer local deployment** for reproducibility-critical tasks when model capability permits.
3. **Implement provenance hashing** to enable post-hoc verification and auditing.
4. **Report full model specifications**: model ID, version, temperature, seed, and any API-specific parameters.
5. **Use majority voting** when local deployment is not feasible, accepting the cost trade-off for improved stability.

6. **Distinguish accuracy from reproducibility** in evaluations—a model can be accurate on average but unreliable for any single run.

6. Conclusion

We have shown that LLM non-determinism is not a theoretical concern but a measurable, quantifiable phenomenon that directly threatens the reproducibility of evidence synthesis. Our experiment comprised 36,000 LLM calls across 120 runs involving six models (three local, three cloud-based APIs), consuming 207 hours of inference time, 55.8 million tokens, and \$69.66 in API costs. This constitutes, to our knowledge, the most comprehensive reproducibility evaluation of LLMs for evidence synthesis to date. Four findings carry immediate practical implications.

First, local deployment achieves perfect reproducibility: all three open-weight models (LLaMA 3 8B, Mistral 7B, Gemma 2 9B) produced bit-identical outputs across all 10 runs in both screening and extraction, demonstrating that deterministic LLM inference is technically achievable on fixed hardware configurations. Second, all three cloud-based APIs (Claude Sonnet 4.5, GPT-4.1, Gemini 2.5 Pro) produce variable outputs even under "deterministic" settings, with extraction tasks amplifying this instability by 13–37× (measured as the flip-rate ratio between extraction and screening): while prior work reports 94–97% reproducibility using per-field metrics and 2–3 runs, our whole-output analysis across 10 runs reveals that only 5–20% of extraction outputs are truly identical. Third, semantic equivalence analysis—including BERTScore embedding similarity ($F_1 \geq 0.997$)—reveals a paradox: models produce near-identical metadata text across runs, yet extract different numeric values and different numbers of effect estimates, driving hash-based EMR to 5–20%. Fourth, meta-analytic propagation analysis reveals that 5–21% of articles produce a different number of effect estimates across runs depending on the API provider, causing the composition of a meta-analysis to depend on which LLM run generated its inputs.

These results carry a clear message: the current standard of reporting LLM-assisted evidence synthesis results from a single, undocumented run is scientifically insufficient. As LLMs become embedded in systematic reviews and meta-analyses, the research community must adopt reproducibility-aware practices—multiple repetitions, provenance tracking, and transparent reporting of variation—or risk building an evidence base on foundations that shift each time the analysis is re-run.

Acknowledgments. The authors thank the PubMed/NCBI E-utilities service for programmatic access to the biomedical literature.

Funding Statement. This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. API costs for Claude, Gemini, and GPT-4.1 experiments were funded by the authors.

Competing Interests. The authors declare no competing interests.

Data Availability Statement. All code, data, prompts, schemas, raw outputs, and run provenance records are publicly available at https://github.com/Roverlucas/llm-evidence-synthesis-reproducibility. API keys are excluded from the repository. The 500 PubMed abstract identifiers, gold standard labels, and complete analysis scripts are included to enable full reproduction of the reported results.

Author Contributions. L.R. conceived and designed the study, developed the software pipeline, conducted all experiments, performed the statistical analysis, and wrote the manuscript. Y.S.T. supervised the research, provided critical feedback on the methodology and interpretation of results, and reviewed the manuscript.

References

1. Higgins JPT, Thomas J, Chandler J, et al. Cochrane Handbook for Systematic Reviews of Interventions. 2nd. Wiley, 2019.
2. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 Statement: An Updated Guideline for Reporting Systematic Reviews. BMJ 2021;372:n71.
3. Borah R, Brown AW, Capers PL, and Kaiser KA. Analysis of the Time and Workers Needed to Conduct Systematic Reviews. BMJ Open 2017;7:e012545.
4. Achiam J, Adler S, Agarwal S, Ahmad L, Akkaya I, et al. GPT-4 Technical Report. arXiv preprint 2023. arXiv:2303.08774.
5. Anthropic. The Claude 3 Model Family: Opus, Sonnet, Haiku. Tech. rep. Anthropic, 2024.

6. Comanici G, Bieber E, Schaekermann M, Pasupat I, Sachdeva N, et al. Gemini 2.5: Pushing the Frontier with Advanced Reasoning, Multimodality, Long Context, and Agentic Capabilities. arXiv preprint 2025. arXiv:2507.06261.
7. Oami T, Okada Y, and Nakada Ta. Performance of a Large Language Model in Screening Citations. JAMA Network Open 2024;7:e2420496.
8. Khraisha Q et al. Can Large Language Models Replace Humans in Systematic Reviews? Research Synthesis Methods 2024;15:707–22.
9. Gartlehner G, Kahwati L, Hilscher R, et al. Data Extraction for Evidence Synthesis Using a Large Language Model: A Proof-of-Concept Study. Research Synthesis Methods 2024;15:576–89.
10. Jensen MK et al. ChatGPT-4o Can Serve as the Second Rater for Data Extraction in Systematic Reviews. PLOS ONE 2025;20:e0313401.
11. Wang Z, Cao L, Danek B, Jin Q, Lu Z, and Sun J. Accelerating Clinical Evidence Synthesis with Large Language Models. npj Digital Medicine 2025;8:509.
12. Li D, Jiang N, Huang K, Tu R, Ouyang S, Yu H, et al. Streamlining Evidence-Based Clinical Recommendations with Large Language Models. npj Digital Medicine 2025;8:2273.
13. Li J et al. Enhancing Systematic Literature Reviews with Generative AI. Journal of the American Medical Informatics Association 2025;32:789–801.
14. Scherbakov D et al. The Emergence of Large Language Models as Tools in Literature Reviews. Journal of the American Medical Informatics Association 2025;32:ocaf063.
15. Matsui K, Utsumi T, Aoki Y, Maruki T, Takeshima M, and Takaesu Y. Human-Comparable Sensitivity of Large Language Models in Identifying Eligible Studies Through Title and Abstract Screening. Journal of Medical Internet Research 2024;26:e52758.
16. Atil B, Aykent S, Chittams A, et al. Non-Determinism of "Deterministic" LLM Settings. arXiv preprint 2024. arXiv:2408.04667.
17. Klishevich E, Denisov-Blanch Y, Obstbaum S, Ciobanu I, and Kosinski M. Measuring Determinism in Large Language Models for Software Code Review. arXiv preprint 2025. arXiv:2502.20747.
18. Fu T, Martínez G, Conde J, et al. Beyond Reproducibility: Token Probabilities Expose Large Language Model Nondeterminism. arXiv preprint 2026. arXiv:2601.06118.
19. Karelin G, Nakajima K, Soto-Astorga EF, Carr E, James M, and Froese T. Indeterminism in Large Language Models: An Unintentional Step Toward Open-Ended Intelligence. Preprint submitted to Artificial Life 2024. PhilSci Archive 26807.
20. Rover L. Hidden Non-Determinism in Large Language Model APIs: A Lightweight Provenance Protocol for Reproducible Generative AI Research. Journal of Artificial Intelligence Research 2026. submitted.
21. Hutson M. Artificial Intelligence Faces Reproducibility Crisis. Science 2018;359:725–6.
22. Haibe-Kains B, Adam GA, Hosny A, et al. Transparency and Reproducibility in Artificial Intelligence. Nature 2020;586:14–6.
23. Kapoor S and Narayanan A. Leakage and the Reproducibility Crisis in Machine-Learning-Based Science. Patterns 2023;4:100804.
24. Han H. Challenges of Reproducible AI in Biomedical Data Science. BMC Medical Genomics 2025;18:2072.
25. Desai S et al. What Is Reproducibility in Artificial Intelligence and Machine Learning Research? AI Magazine 2025;46:e70004.
26. Semmelrock H, Ross-Hellauer T, Kopeinik S, et al. Reproducibility in Machine-Learning-Based Research: Overview, Barriers, and Drivers. AI Magazine 2025;46:e70002.
27. Angermeir F, Amougou M, Kreitz M, et al. Reflections on the Reproducibility of Commercial LLM Performance in Empirical Software Engineering Studies. In: arXiv:2510.25506. 2025.
28. Yu B. After Computational Reproducibility: Scientific Reproducibility and Trustworthy AI. Harvard Data Science Review 2024;6.
29. Longpre S, Mahari R, Chen A, et al. A Large-Scale Audit of Dataset Licensing and Attribution in AI. Nature Machine Intelligence 2024;6:878.
30. Liang P, Rajani N, Yang D, et al. Systematic Analysis of 32,111 AI Model Cards Characterizes Documentation Practice. Nature Machine Intelligence 2024;6:857.
31. Bommasani R, Klyman K, Longpre S, et al. The Foundation Model Transparency Index v1.1. Transactions on Machine Learning Research 2024.

32. Leo S, Crusoe MR, Rodríguez-Navas L, et al. Recording Provenance of Workflow Runs with RO-Crate. PLOS ONE 2024;19:e0309210.

33. Kale A, Nguyen T, Harris FC, Li C, Zhang J, and Ma X. Provenance Documentation to Enable Explainable and Trustworthy AI: A Literature Review. Data Intelligence 2023;5:139–62.

34. Longpre S et al. Data Authenticity, Consent, and Provenance for AI Are All Broken. In: *Proceedings of ICML 2024 (Position Paper)*. arXiv:2404.12691. 2024.

35. Yu W, Xu R, Ye T, et al. Estimates of Global Mortality Burden Associated with Short-Term $PM_{2.5}$ Exposure. The Lancet Planetary Health 2024;8:e146–e155.

36. Iyer HS, Karasaki S, Yi L, Hswen Y, James P, and VoPham T. Harnessing Geospatial Artificial Intelligence (GeoAI) for Environmental Epidemiology: A Narrative Review. Current Environmental Health Reports 2025;12:497.

37. Yu M, Fang M, Tian Z, Wang B, and Walker D. Artificial Intelligence and Machine Learning for Environmental Health Study. Environment & Health 2025;3:1115–6.

38. Dubey A, Jauhri A, Pandey A, Kadian A, Al-Dahle A, et al. The LLaMA 3 Herd of Models. arXiv preprint 2024. arXiv:2407.21783.

39. Jiang AQ, Sablayrolles A, Mensch A, Bamford C, Chaplot DS, et al. Mistral 7B. arXiv preprint 2023. arXiv:2310.06825.

40. Gemma Team, Mesnard T, Hardin C, Dadashi R, Bhupatiraju S, et al. Gemma 2: Improving Open Language Models at a Practical Size. arXiv preprint 2024. arXiv:2408.00118.